Travis Brown

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OBJECTIVE

Seeking an embedded software and firmware engineering internship for Summer 2027.

EDUCATION

Rochester Institute of Technology Expected: December 2027
Master of Science and Bachelor of Science in Computer Science **GPA: 3.96**

SKILLS

Languages: C, C++, Python, Rust, JavaScript, MIPS Assembly

Embedded: STM32 / ESP32 microcontrollers, FreeRTOS / ThreadX RTOS, CAN/CANopen, Modbus, MQTT, SPI, I2C, UART, PWM, ADC, PID control, SITL simulation, OpenOCD, J-Link Segger, logic analyzer, oscilloscope

Tools: Git, GitHub, GitLab, Linux, macOS, KiCad, Onshape, STM32 CubeMX / CubeIDE

EXPERIENCE

Embedded Software Intern
GE Vernova May 2026 – Present

- Built a full I/O subsystem from scratch for industrial radios on STM32F7, supporting digital in/out, analog input, PWM output, and interrupt-driven edge detection across three configurable I/O ports.
- Implemented Modbus TCP, RTU, and ASCII servers with a register map exposing diagnostics, I/O states, and network health.
- Integrated MQTT and MQTT-SN support over TCP, TLS, and UDP to the radio's firmware, with user-uploadable certificates for private brokers, enabling remote diagnostics and fleet-wide configuration.
- Diagnosed servo jitter blocking a customer demo, ruling out signal integrity and timing via oscilloscope and Saleae before tracing the fault to a defective servo, enabling the demo to proceed on schedule.

Software QA Automation Intern
Lockheed Martin May 2025 – August 2025

- Developed a Python tool automating over 60% of QA's manual merge request (MR) checklist across 100+ repos, cutting review time by 75% (20 min → 5 min); added as a required GitLab CI check after several QA feedback rounds.
- Authored checks for secrets, commented-out code, dependency versions, README contents, and MR/Jira ticket validity.
- Published results to a GitLab Pages report with color-coded status, recommendations, and links for instant developer feedback.

Embedded Software Engineer
RIT Electric Vehicle Team August 2023 – Present

- Engineered C++ firmware for an STM32F446 vehicle control unit (VCU) of an electric motorcycle, following cross-team ISO 26262 principles (risk analysis, safety requirements, safe states, testing); integrated the controls team's Simulink state machine at 300 Hz.
- Structured the VCU's multithreaded ThreadX design with a timer-triggered model thread, health monitoring, interrupt-fed CAN queues, mutexes, and event flags for drive-by-wire control and bike-wide fault handling.
- Connected the VCU to the motor controller, BMS, and handlebar interface board (HIB) over CAN and to the low-voltage subsystem (LVSS) over CANopen; designed the VCU's CANopen dictionary and message formats, co-designing LVSS/HIB payloads.
- Delivered a UART/NMEA 0183 GNSS driver for the Gateway Utility Board with checksums, message filtering, and configurable rate.
- Expanded our hardware-agnostic library with STM32F446 peripheral drivers (PWM, ADC, SPI, timers) using STM32 HAL.

PROJECTS

Autonomous Fixed-Wing Flight Controller
Personal Project June 2026 – Present

- Programmed a custom ESP32S3 based flight controller in C using FreeRTOS, implementing autonomous PID stabilization (roll and pitch), sensor fusion, GPS waypoint sequencing, and manual override.
- Created attitude and navigation SITL simulators; validated roll, pitch, and heading PID gains across 80 regression tests covering step responses, noise sweeps, and waypoint missions, ahead of autonomous flight testing.
- Designed a flight controller PCB in KiCad with dual onboard IMUs, onboard barometer, SPI/I2C/UART sensor buses, eight PWM outputs, Schottky reverse-current protection on the 5 V input, and separate servo, logic, and sensor supply rails.

Handheld Chip-8 Emulator
Personal Project June 2025 – December 2025

- Wrote a Chip-8 emulator in Rust on a Raspberry Pi, including an SPI display driver for the SSD1309 OLED.
- Hand-soldered a keypad PCB designed in KiCad; modeled and 3D-printed an enclosure in Onshape.
- Extended the emulator with configurable cycle speed and per-ROM quirk flags; wrote a game-selection ROM in Octo assembly.